FACTORING THROUGH A GOOD MAP WITH VANISHING HOM

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Allowed standard prerequisites (editorial).

- Complex representation. derived-lemma-bounded-derived (cross-chapter, named). Represents f by a map of bounded complexes.

Source reference. Original reference 05RW: focused-seeds/geometry/S9-eqv-good-map-zero/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with enough negative objects. Let $f : X \to X'$ be a morphism of $D^b(\mathcal{A})$. Let $b \in \mathbf{Z}$ such that $H^i(X) = 0$ for $i > b$ and $H^i(X') = 0$ for $i \geq b$. Then there exists a map $N[-b] \to X$ such that the induced map $N \to H^b(X)$ is surjective, such that $\mathrm{Hom}(H^b(X), N) = 0$, and such that the composition $N[-b] \to X \to X'$ is zero.*

Proof. We can represent f by a map $f^\bullet : A^\bullet \to B^\bullet$ of bounded complexes of objects of $\mathcal{A}$, see for example Derived Categories, Lemma 05RW. Consider the object

$$C = \mathrm{Ker}(A^b \to A^{b+1}) \times_{\mathrm{Ker}(B^b \to B^{b+1})} B^{b-1}$$

of $\mathcal{A}$. Since $H^b(B^\bullet) = 0$ we see that $C \to H^b(A^\bullet)$ is surjective. On the other hand, the map $C \to A^b \to B^b$ is the same as the map $C \to B^{b-1} \to B^b$ and hence the composition $C[-b] \to X \to X'$ is zero. Since $\mathcal{A}$ has enough negative objects, we can find an object N which has a surjection $N \to C \oplus H^b(X)$ such that $\mathrm{Hom}(C \oplus H^b(X), N) = 0$. Then N together with the map $N[-b] \to X$ is a solution to the problem posed by the lemma. □

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